XRPL Validator — Daily Engineering Report · 2026-09-15

Period: 2026-09-15 00:00 to 24:00 US Eastern (2026-09-15 04:00 to 2026-09-16 04:00 UTC); ledgers #106,994,221 to #107,016,556. Every count from the ledger and the validator's logs stops at #107,016,556. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 43 engine receipts recorded (3 fixed, 0 explained, 40 without a recorded fix).
- **Changes:** 2 deployments (cycles None, 134), 1 commit merged into main, 1 engine finding fixed (F290).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Ledgers with a receipt in the period (#106,994,221–#107,016,556), by step:

Step	Ledgers with a receipt
Step 1 · native engine	43
Step 2 · records	0
Step 3 · order	0

Step	Ledgers with a receipt
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#106,994,584	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-02	—	#106,994,585	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-03	—	#106,994,627	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-04	—	#106,994,813	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-05	—	#106,994,815	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-06	—	#106,994,887	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-07	—	#106,994,919	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-08	—	#106,994,921	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-09	—	#106,994,962	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-10	—	#106,995,247	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-11	—	#106,995,761	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-12	—	#106,995,763	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-13	—	#106,995,769	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-14	—	#106,995,774	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-15	—	#106,995,777	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-16	—	#106,996,075	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-17	—	#106,997,573	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-18	—	#106,997,785	Step 1 · native engine	2 object(s) differed in content	no disposition recorded
R-19	—	#106,997,786	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded

ID	Time	Ledger(s)	Step	Description	Disposition
R-20	—	#106,997,968	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-21	—	#106,997,994	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-22	—	#106,997,995	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-23	—	#106,997,996	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-24	—	#106,997,997	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-25	—	#106,997,998	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-26	—	#106,997,999	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-27	—	#106,999,022	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-28	—	#106,999,024	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-29	—	#106,999,115	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-30	—	#106,999,138	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-31	—	#106,999,139	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-32	—	#106,999,141	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-33	—	#106,999,205	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-34	—	#106,999,444	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-35	—	#106,999,445	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-36	—	#106,999,572	Step 1 · native engine	4 change(s) of ours the network did not make	fixed: F292, commit 7bd0c74, deployed in cycle 135 (2026-09-17 15:08 UTC)
R-37	—	#106,999,761	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-38	—	#106,999,762	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-39	—	#107,000,904	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-40	—	#107,002,363	Step 1 · native engine	4 change(s) of ours the network did not make	fixed: F292, commit 7bd0c74, deployed in cycle 135 (2026-09-17 15:08 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-15-01, INC-01 is INC-2026-09-15-01. Times are not shown where the reference node no longer holds the ledger.

3 more receipts (#107,004,340–#107,009,438) are in the record and not listed individually; their dispositions: 1 fixed, 2 no disposition recorded.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
None	`c089330`	04:23:36 UTC	n/a	n/a	n/a	#9 from 04:23 UTC: ended early: superseded by the next deployment (2026-09-15 10:42 UTC)
134	`9718018`	10:42:00 UTC	n/a	n/a	401 matched, 0 mismatched	#10 from 10:42 UTC: ended early: superseded by the next deployment (2026-09-17 15:08 UTC)

Cycle None delivered: `c089330` test(xrpl-node): drop a stray brace appended after the F282 vector (gate131 SUITES build error)

Cycle 134 delivered: `9718018` fix(xrpl-ledger): F290 — after a pool-served pass the stream steps the book only when the pool would NOT offer again

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
04:59 UTC	xrpl-validator	`9718018`	fix(xrpl-ledger): F290 — after a pool-served pass the stream steps the book only when the pool would NOT offer again

Findings fixed

- **fix(xrpl-ledger): F290 — after a pool-served pass the stream steps the book only when the pool would NOT offer again** (`9718018` , xrpl-validator) #106992486 1E418A4F5724 (and siblings 106992487, 106992529): rDireAucG's partial self-payment of a sentinel XRP amount for 226662 ATM through the ATM/XRP pool. rippled's next iteration asks the pool again on its post-slice balances: with an offer (changeSpotPriceQuality succeeds — 459164192.94 ATM for 2395603301 drops, priced between the spot and the tip) `tryAMM` executes it and `execOffer(tip)` fails `*ofrQ != …

Docs and operations

- **flow: first calibration of the port against the payment vectors — 20 → 42 of 65** (`16c228f` , xrpl-validator)
- **flow: second calibration round — 42 → 60 of 65 payment vectors** (`17b9728` , xrpl-validator)

- **flow: third calibration round — 60 → 64 of 65 (xrpLiquid hooks the full balance before the reserve; in-flow offer deletions go through adjustOwnerCountHook)** (`5f1570d`, xrpl-validator)
- **flow: 65/65 — the ported Payment flow matches every pinned payment vector** (`ccc7b94`, xrpl-validator)
- **flow: OfferCreate crossing dispatch — CreateOffer::flowCross port behind XRPL_FLOW_ENGINE=port (first run: 79 of 106 offer vectors)** (`7b73eaa`, xrpl-validator)
- **flow: crossing calibration — 79 → 95 of 106 offer vectors (maxOffer carries the spot quality; tip() applies the crossing's qualityThreshold)** (`d9abcef`, xrpl-validator)

Independence program

No independence record for UTC day 2026-09-15; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- BatchV1_1 gained majority — ledger 107,003,649, 2026-09-15 14:06 UTC

Amendment	Majority since	Activation due
BatchV1_1	2026-09-15 14:06 UTC	2026-09-29 14:06 UTC

Amendments enabled on mainnet: 94.

Open items

- Receipt R-01 (#106,994,584): open, under investigation.
- Receipt R-02 (#106,994,585): open, under investigation.
- Receipt R-03 (#106,994,627): open, under investigation.
- Receipt R-04 (#106,994,813): open, under investigation.
- Receipt R-05 (#106,994,815): open, under investigation.
- Receipt R-06 (#106,994,887): open, under investigation.
- Receipt R-07 (#106,994,919): open, under investigation.

- Receipt R-08 (#106,994,921): open, under investigation.
- Receipt R-09 (#106,994,962): open, under investigation.
- Receipt R-10 (#106,995,247): open, under investigation.
- Receipt R-11 (#106,995,761): open, under investigation.
- Receipt R-12 (#106,995,763): open, under investigation.
- Receipt R-13 (#106,995,769): open, under investigation.
- Receipt R-14 (#106,995,774): open, under investigation.
- Receipt R-15 (#106,995,777): open, under investigation.
- Receipt R-16 (#106,996,075): open, under investigation.
- Receipt R-17 (#106,997,573): open, under investigation.
- Receipt R-18 (#106,997,785): open, under investigation.
- Receipt R-19 (#106,997,786): open, under investigation.
- Receipt R-20 (#106,997,968): open, under investigation.
- Receipt R-21 (#106,997,994): open, under investigation.
- Receipt R-22 (#106,997,995): open, under investigation.
- Receipt R-23 (#106,997,996): open, under investigation.
- Receipt R-24 (#106,997,997): open, under investigation.
- Receipt R-25 (#106,997,998): open, under investigation.
- Receipt R-26 (#106,997,999): open, under investigation.
- Receipt R-27 (#106,999,022): open, under investigation.
- Receipt R-28 (#106,999,024): open, under investigation.
- Receipt R-29 (#106,999,115): open, under investigation.
- Receipt R-30 (#106,999,138): open, under investigation.
- Receipt R-31 (#106,999,139): open, under investigation.
- Receipt R-32 (#106,999,141): open, under investigation.
- Receipt R-33 (#106,999,205): open, under investigation.
- Receipt R-34 (#106,999,444): open, under investigation.
- Receipt R-35 (#106,999,445): open, under investigation.
- Receipt R-37 (#106,999,761): open, under investigation.
- Receipt R-38 (#106,999,762): open, under investigation.
- Receipt R-39 (#107,000,904): open, under investigation.
- Receipt R-41 (#107,004,340): open, under investigation.
- Receipt R-42 (#107,007,895): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Aggregate spend caps for session keys** (accounts) Add a new owner-reserved singleton ledger object AccountPolicy (ltACCOUNT_POLICY, keylet::accountPolicy(account)) with sfPolicySpendLimit, sfPolicyWin
- **Credential-gated fee and reserve sponsorship** (compliance) Add an sfAcceptedCredentials field to the Sponsor ledger object/transaction (reusing the STArray<{Issuer,CredentialType}> encoding already defined for
- **On-ledger validator reputation from HardenedValidations** (consensus) Add a new per-validator singleton ledger object ValidatorReputation (ltVALIDATOR_REPUTATION, keylet::validatorRep(masterPubKey)) holding sfAgreementSc

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-15 04:25	—	xrpl-validator	`16c228f`	flow: first calibration of the port against the payment vectors — 20 → 42 of 65
09-15 04:19	09-15 04:59	xrpl-validator	`9718018`	fix(xrpl-ledger): F290 — after a pool-served pass the stream steps the book only when the pool would NOT offer again
09-15 05:09	—	xrpl-validator	`17b9728`	flow: second calibration round — 42 → 60 of 65 payment vectors
09-15 05:11	—	xrpl-validator	`5f1570d`	flow: third calibration round — 60 → 64 of 65 (xrpLiquid hooks the full balance before the reserve; in-flow offer deletions go through adjustOwnerCountHook)
09-15 05:14	—	xrpl-validator	`ccc7b94`	flow: 65/65 — the ported Payment flow matches every pinned payment vector
09-15 05:18	—	xrpl-validator	`7b73eaa`	flow: OfferCreate crossing dispatch — CreateOffer::flowCross port behind XRPL_FLOW_ENGINE=port (first run: 79 of 106 offer vectors)
09-15 05:20	—	xrpl-validator	`d9abcef`	flow: crossing calibration — 79 → 95 of 106 offer vectors (maxOffer carries the spot quality; tip() applies the crossing's qualityThreshold)

Times in UTC. A commit appears on the day it was committed to main or merged into it.